

Dr Thomas Frost

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PROFILE

Machine learning researcher and Oxford-trained physician, currently concluding a PhD in AI-enabled healthcare at UCL. My research focuses on the deployment of reinforcement learning within high-stakes, complex real-world systems. It is my belief that healthcare represents one of the ultimate benchmarks for machine intelligence, with sparse rewards, potentially severe consequences from suboptimal decisions, and very partially-observable states. As part of my PhD, I built a sophisticated dose recommendation system for insulin infusions in intensive care; released [Insulin4RL](#), an open-source dataset for healthcare-based reinforcement learning with realistic data; supported the local development of [FlowEHR](#), a custom hospital platform for internal ML model development and deployment; and supported the evaluation of LLM-generated discharge summaries.

EDUCATION

University College London (UCL) London, UK
PhD in AI-enabled Healthcare Sept 2022 — Expected Oct 2026

- **Thesis:** “Offline reinforcement learning for clinical data with natural timings.”
- Designed and built an offline reinforcement learning system for ICU drug dose recommendation in the context of delayed rewards, confounding, and high-stakes decision-making.
- Created preprocessing pipelines for noisy, high-frequency time-series data across multiple clinical sites to ensure model robustness and safety in critical environments.
- Partnered with Microsoft and UCLH to help refine FlowEHR (<https://www.safehr-data.org/our-tools>), an open-source MLOps project for deployment of ML models into clinical workflows.
- Served on the clinical evaluation team for LLM-generated discharge summaries, defining acceptance criteria and identifying hallucination patterns in medical text.

Royal College of Emergency Medicine London, UK
MRCEM (Member of the Royal College of Emergency Medicine) Dec 2019 — Jan 2024

University College London (UCL) London, UK
MRes (Master of Research) in AI-enabled Healthcare Sept 2021 — Sept 2022

University of Oxford Oxford, UK
BM BCh (Bachelor of Medicine, Bachelor of Surgery) Oct 2012 — Jul 2018

University of Oxford Oxford, UK
MA in Medical Sciences, First Class Oct 2014 — Jul 2015

WORK EXPERIENCE

Emergency Medicine Clinician 2018 — Present
NHS Scotland; St George's / Homerton Foundation Trusts Scotland & London, UK

- Delivering frontline care in high-pressure environments, routinely making safety-critical decisions with incomplete data.
- Coordination of multidisciplinary teams for management of trauma and resuscitation for unwell patients.

TECHNICAL & CLINICAL SKILLS

- **Core:** Python, PyTorch, and data science suite (NumPy/Pandas/Polars/SciPy).
- **Deep ML:** Offline Reinforcement Learning, Time-Series Forecasting, Clinical Decision Support.
- **Cloud & Infrastructure:** AWS Compute (EC2/S3 for research workloads), FlowEHR (TRE User/Validator).
- **Clinical Products:** Human-in-the-loop Evaluation, Clinical Safety Metrics, Medical Data Privacy.

AWARDS AND RECOGNITION

- **UCL Three-Minute Thesis:** University finalist, communicating complex AI concepts to non-technical audiences.
- **UCL Centre of Digital Innovation Scholar:** Awarded £5,000 AWS research credits + cloud computing mentorship.
- **UKRI Studentship:** Full PhD stipend + annual research budget for outstanding research potential.
- **Lord Crewe Scholar (Oxford):** Awarded for academic distinction.

PUBLICATIONS

- Frost, T., & Harris, S. (2026). Insulin4RL: Real-time insulin infusions for offline reinforcement learning. PhysioNet. <https://doi.org/10.13026/swen-q904>
- Frost, Thomas, and Steve Harris. “Insulin4RL: Real-time insulin management in the intensive care unit for offline reinforcement learning.” arXiv preprint arXiv:2606.19481 (2026).
- Frost, T., Vaidya, H., & Harris, S. (2026). The hidden risks of temporal resampling in clinical reinforcement learning. arXiv preprint arXiv:2602.06603.
- Frost, T., Li, K., & Harris, S. (2025). Robust real-time mortality prediction in the intensive care unit using temporal difference learning. In Proceedings of the 4th Machine Learning for Health Symposium (pp. 350–363). PMLR.
- Ellershaw, S., Tomlinson, C., Burton, O. E., Frost, T., ... & Dobson, R. (2024). Automated generation of hospital discharge summaries using clinical guidelines and large language models. In AAAI 2024 Spring Symposium on Clinical Foundation Models.
- Collins, D. R., Jobanputra, K., Frost, T., ... & Heneghan, C. (2017). Cardiovascular disease risk and prevention amongst Syrian refugees: mixed methods study of Médecins Sans Frontières programme in Jordan. *Conflict and Health*, 11(1), 14.
- Frost, T. D., Sinha, D., & Gilbert, B. J. (2014). Should assisted dying be legalised?. *Philosophy, Ethics, and Humanities in Medicine*, 9(1), 3.